

Tommy Orok

Python Backend Engineer | AI/ML Integration | AWS Deployment | SaaS Platform & API Development

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SUMMARY

I design and build Python backends and data pipelines that turn raw inputs into reliable insights, with a focus on production deployment on AWS. A standout project is a machine learning driven mean reversion system developed in Python with PyTorch on AWS, where I built end-to-end data pipelines ingesting historical market data into S3, engineered features such as z scores, realized volatility, volume anomalies, and moving average distance, and implemented walk forward validation to avoid look ahead bias; the models are deployed as real-time signals via SageMaker endpoints and monitored with CloudWatch and SageMaker Model Monitor. Another key achievement is the Tommy Planner backend, a Mongo-backed API for a recursive planning tool, with an in-memory PlanStore and a DataAdapter connecting optimistic client updates to serverless endpoints; the system supports deep nesting, subplans, references, cycles, and current-topic designations, and maps patterns to production Python backend work like API design and data processing. During a LawNearMe internship I built a web scraper that pulled lawyer names and emails and automated outreach, cutting daily tasks from a full day to under an hour while maintaining compliance and gaining Python automation experience deploying lightweight tooling on cloud infrastructure. I share SaaS Rationalization's goal of turning software usage data into actionable automation, and I am ready to contribute to building reliable, secure, and scalable backend services on AWS with ML model integrations and external API connections to power smart recommendations across customer workloads.

SKILLS

- Python backend development
- API design and development
- AI/ML model integration
- Data pipeline engineering
- AWS deployment and operations
- Technical leadership and collaboration
- Mentor engineers
- Web scraping and automation

EXPERIENCE

AI Consultant | Kacha Inc

August 2023 – Present

Atlanta, Ga

- Implemented end-to-end backend and API development by drafting plans for schema changes, API design, and migrations with AI-driven reviews to deliver production-ready code across server, API, and data layers.
- Engineered schema changes and migration strategies as part of building production APIs, ensuring backward compatibility and consistent data models.
- Applied defense-in-depth validation across API, frontend, and database layers and enforced modular coding practices (limits on file size, no duplicated logic) to boost reliability and maintainability in an evolving project.

Software Consultant | Ivy Kode

2019 – 2025

Atlanta, Ga

- Led hands-on Python backend development and modernization on client projects, wrote code 95% of the time, and migrated legacy systems to modern stacks.
- Delivered production-grade Python backend services and debugging support across client engagements, translating diverse requirements into robust backend solutions.
- Led the technical direction on specialist/project-based engagements, translating client requirements into architecture and delivering outcomes with autonomy in fast-changing environments.

Software Developer | 21st Century Realty

October 2017 – May 2020

Austell, Ga

- Developed and maintained backend services and APIs to support end-to-end checkout, inventory management, and order fulfillment, coordinating session context, carts, discounts, and idempotent checkout flows.
- Implemented authentication and authorization mechanisms using Auth.js sessions, JWTs, and Argon2id hashing, while leveraging MongoDB as the data store for users and permissions.
- Deployed and maintained serverless backend components on a cloud platform (Vercel), implementing Redis-backed distributed locks to coordinate checkout and session state for reliability and scalability.

Head Systems Engineer | Gryphus Trading

December 2015 – August 2019

New York, New York

- Led design and development of ML-driven mean-reversion trading systems using Python, PyTorch, and AWS; built end-to-end data pipelines, feature engineering, and production endpoints for AI-powered signals.
- Deployed and operated AWS-based ML infrastructure for model training, inference, and monitoring (SageMaker, S3, CloudWatch), enabling scalable, production-grade AI capabilities.
- Architected and implemented robust data processing and validation workflows, including historical data ingestion, feature extraction, walk-forward validation, and automated retraining triggers to maintain model performance.

EDUCATION

Bachelor's Degree, Computer Science | Columbia University, New York, New York

August 2012 – May 2016